

DATA ANALYTICS PRACTICE SERIES

Apple Sales Analytics

Data Visualization Workbook

Beginner to Advanced

Project-Based Data Analytics Practice

10 Blocks • 50 Questions • Portfolio Ready

PART 03 OF THE SERIES

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NOTE

Each block progresses from foundational recall (Easy) through applied reasoning (Medium) to a Business Challenge (Hard) rooted in the real Apple Sales dataset — category.csv, products.csv, sales.csv, stores.csv, and warranty.csv.

DATASET AT A GLANCE

The Apple Sales Dataset

category.csv	10 rows	category_id, category_name
products.csv	64 rows	product_id, product_name, category_id (FK), launch_date, price
sales.csv	1,040,191 rows	sale_id, sale_date, store_id (FK), product_id (FK), quantity
stores.csv	73 rows	store_id, store_name, city, country
warranty.csv	30,836 rows	claim_id, claim_date, sale_id (FK), claim_status

WATCH FOR

There is no revenue column anywhere in this dataset — every revenue question requires joining sales.csv → products.csv and computing quantity × price. quantity ranges only 1–6, with 94% of all transactions single-unit. sale_date is ISO format (YYYY-MM-DD); launch_date and claim_date are DD-MM-YYYY. The dataset spans January 2019 – August 2024, and 2024 is a partial year. Referential integrity is clean across all five tables.

PART THREE

Data Visualization Workbook

Ten blocks. Fifty questions. From choosing the right chart to a Power-BI-ready dashboard storyboard, built entirely on real Apple sales data.

Visualization Fundamentals

CHOOSING THE RIGHT CHART · FIGURE & AXES · TITLES & LABELS · LEGENDS · GRIDLINES · COLOR · FIGURE SIZE

EASY Q1

Plot the distribution of price from products.csv, choosing a chart type suited to a continuous numeric variable.

EASY Q2

Join products.csv with category.csv and plot the count of products per category_name, choosing a chart suited to categorical data.

MEDIUM Q3

Join sales.csv, products.csv, and category.csv, then visualize total units sold per category_name using a sorted, gridded bar chart.

MEDIUM Q4

Visualize the number of Apple stores per country from stores.csv, using a defined figsize, rotated axis labels, and a descriptive title.

HARD – BUSINESS CHALLENGE Q5

For five real requests — a revenue trend, a price-spread comparison across categories, a category revenue share, a price-vs-units relationship, and a warranty claim_status breakdown — choose and justify the correct chart type for each, and explain which one you would lead with in a CEO meeting.

Bar Charts

BAR · BARH · SORTING · DATA LABELS / ANNOTATIONS

EASY Q1

Join sales.csv and products.csv to derive revenue (quantity × price), then plot the Top 10 products by revenue as a sorted bar chart.

EASY Q2

Join sales.csv, products.csv, and category.csv to plot revenue by category_name across all 10 categories, with value labels above each bar.

MEDIUM Q3

Join sales.csv and stores.csv to plot the Top 15 stores by units sold as a horizontal bar chart, and explain why the Top 15 cutoff matters.

MEDIUM Q4

Join sales.csv, products.csv, and stores.csv to plot the Top 15 countries by revenue, highlighting the leading country in a distinct color.

HARD – BUSINESS CHALLENGE Q5

Build a two-panel bar report showing Revenue by Category and the Top 10 Stores by Revenue, annotate the top performer in each panel, and summarize what the report tells management.

BLOCK 03

Line Charts

SALES TRENDS · REVENUE TRENDS · MULTIPLE LINES · MARKERS

EASY Q1

Parse `sale_date` and plot the monthly units-sold trend across the full `sales.csv` dataset.

EASY Q2

Join `sales.csv` and `products.csv` to derive revenue, then plot the monthly revenue trend with markers at each point.

MEDIUM Q3

Filter `sales.csv` to one Nov-Dec holiday window and plot the daily quantity trend, marking the single highest-volume day.

MEDIUM Q4

Compare monthly revenue trends across 2019–2023 using both `plt.plot()` and `sns.lineplot()`, and explain why 2024 should be excluded from the comparison.

HARD – BUSINESS CHALLENGE Q5

Combine the monthly units and revenue trends into one two-panel figure, label the single highest-revenue month, and explain what the seasonal pattern suggests for inventory planning.

Distribution Analysis

HISTOGRAM · KDE · COUNTPLOT · DISTRIBUTION SHAPE

EASY Q1

Plot a histogram with a KDE overlay of price from products.csv, and describe where most of Apple's catalog sits.

EASY Q2

Join sales.csv and products.csv to derive a per-transaction revenue value, then plot its distribution and describe its skew.

MEDIUM Q3

Plot the distribution of quantity from sales.csv as a countplot rather than a histogram, and report what share of orders are single-unit.

MEDIUM Q4

Plot the count of warranty claims per claim_status from warranty.csv, and explain whether a pie chart would be a better choice here.

HARD – BUSINESS CHALLENGE Q5

Combine the price, quantity, and claim_status distributions into a single diagnostic figure and flag anything operationally unusual.

Relationship Analysis

SCATTER PLOT · REGRESSION LINE · CORRELATION INTUITION

EASY Q1

Aggregate sales.csv to one row per product and plot price against total units sold as a scatter plot.

EASY Q2

Using the same product-level aggregation, plot price against total revenue with a regression line.

MEDIUM Q3

Build a category-level table of total revenue and total warranty claims, then plot the relationship between them across all 10 categories.

MEDIUM Q4

Build a store-level table of total revenue and transaction count, then plot the relationship across all 73 stores and flag any outlier store.

HARD – BUSINESS CHALLENGE Q5

Combine the price-vs-revenue and revenue-vs-transaction-count scatter plots into one figure, annotate the strongest relationship, and recommend where management should focus to grow revenue.

BLOCK 06

Seaborn Business Charts

COUNTPLOT · BARPLOT · LINEPLOT · SCATTERPLOT

EASY Q1

Join products.csv and category.csv, then use `sns.countplot()` to show the number of distinct products per category.

EASY Q2

Filter stores.csv to countries with two or more stores and use `sns.barplot()` to compare average store revenue by country.

MEDIUM Q3

Rebuild the Top 15 Countries by Revenue chart using `sns.barplot()` and compare it against your matplotlib version from Block 02.

MEDIUM Q4

Join warranty.csv through to category.csv and use a seaborn grouped bar chart to compare claim_status counts across categories.

HARD – BUSINESS CHALLENGE Q5

Build a single seaborn dashboard figure combining category revenue and warranty status using one consistent theme, and recommend one action for the support team.

Advanced Seaborn — Outlier & Spread Detection

BOXPLOT · VIOLINPLOT · STRIPLOT · SWARMPLOT

EASY Q1

Join products.csv and category.csv, then use `sns.boxplot()` to compare price spread across all 10 categories.

EASY Q2

Derive per-transaction revenue and use `sns.violinplot()` to compare its spread across the top 5 categories by transaction count.

MEDIUM Q3

Use `sns.stripplot()` or `sns.swarmplot()` to show individual quantity values by category, and explain why this suits a discrete variable better than a boxplot.

MEDIUM Q4

Filter to one country with multiple stores and use `sns.stripplot()` to flag the lowest-performing store by revenue.

HARD — BUSINESS CHALLENGE Q5

Combine a price-by-category boxplot with a `claim_status` swarm/strip view to identify one category worth a closer quality review, and justify your choice.

Correlation & Heatmaps

CORRELATION MATRIX · HEATMAP · PAIRPLOT

EASY Q1

Build a product-level table of price, total units sold, total revenue, and total warranty claims, then render its correlation matrix as an annotated heatmap.

EASY Q2

Build a category-by-year revenue pivot table and render it as a heatmap, noting which category grew or declined most.

MEDIUM Q3

Build a pairplot of price, total quantity, and total revenue at the product level, colored by category.

MEDIUM Q4

Build a store-by-category quantity pivot table for the Top 15 stores and render it as a heatmap to spot any unusually concentrated category mix.

HARD – BUSINESS CHALLENGE Q5

Using the correlation and heatmap findings from this block, recommend three specific KPIs Apple's leadership should track and justify each with a chart.

Business Dashboard Visualizations

DASHBOARD ASSEMBLY · PRE-BI STORYBOARDING

EASY Q1

Build the Top 10 Products by Revenue and Top 15 Stores by Revenue as production-quality bar charts.

EASY Q2

Build the Top 15 Countries by Revenue and the Category Revenue Share as a percentage of total, choosing the right chart for the share-of-whole view.

MEDIUM Q3

Build the monthly revenue trend, monthly units-sold trend, and warranty claims by claim_status as a three-panel figure.

MEDIUM Q4

Build a price-mix boxplot by category and a ranked Top 20 stores by revenue chart, applying one consistent color palette across every chart in this block.

HARD – BUSINESS CHALLENGE Q5

Assemble all 10 dashboard charts from this block into a single subplot grid with one dashboard-level title, ready to hand to a Power BI developer.

Final Visualization Project

FULL EDA NOTEBOOK · EXECUTIVE STORYTELLING · GITHUB-READY DELIVERY

EASY Q1

Structure a notebook with a markdown overview of all 5 tables, their row counts, and how they join.

EASY Q2

Select your 6 strongest charts from Blocks 01–09 and re-render them at high resolution as an executive chart set.

MEDIUM Q3

Write a 2–3 sentence business narrative for each of your 6 executive charts, focused on what it shows and what to do next.

MEDIUM Q4

Re-export every executive chart as a standalone, presentation-ready image file with no code visible.

HARD – BUSINESS CHALLENGE Q5

Combine your executive chart set, business narratives, and dashboard storyboard into one complete report, close it with 3 data-driven actions for retail leadership, and structure the project to be GitHub-README ready.



Data Visualization Workbook Complete

Data Analytics Practice Series — Part 03

- ✓ 10 Blocks
- ✓ 50 Questions
- ✓ Real Apple Sales Dataset
- ✓ Matplotlib + Seaborn
- ✓ Beginner → Advanced
- ✓ Portfolio Ready
- ✓ Power BI Ready
- ✓ Interview Oriented